Employees in the US Nonprofit Sector

by

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Abstract

The nonprofit sector employs roughly 10% of the American workforce, making it the third largest workforce behind the retail and manufacturing sectors. Despite this, relatively little is known about its employees. This paper is the first to use comprehensive administrative tax data, covering the near-universe of workers in the US, to quantify and explain the causes of the nonprofit pay differential. Unconditionally, we find the nonprofit earnings penalty to be 12% relative to for-profit workers. Estimating an "AKM" worker-firm job ladder model, we show that most of the penalty is causal and not driven by selection. We also document considerable heterogeneity across industries, both in terms of earnings premia/penalties and worker selection, and show that nonprofit and for-profit earnings have been converging over time.

* Disclaimer: Any views expressed are those of the authors and not those of the U.S. Census Bureau or any agency of the U.S. federal government. This research was performed at a Federal Statistical Research Data Center under FSRDC Project Number 2459. Results were approved for release by the U.S. Department of the Treasury.

1. Introduction

Despite the fact that the nonprofit sector employs roughly 10% of the American workforce, making it the third largest workforce behind retail and manufacturing[1], relatively little is known about its employees. This project is the first to use administrative data to paint a picture of the U.S. nonprofit sector, document some basic facts about its workers, and quantify and explain the causes of the nonprofit pay differential.

The project began by creating a novel data linkage between the American Community Survey (ACS), the IRS Form 990, and administration earnings data from the Longitudinal Employer Household Dynamics (LEHD) to study the nonprofit labor market. However, in 2024 we gained access to administrative tax data on the universe of US workers from 1999-2023. These data afforded us the opportunity to reassess the nonprofit pay premium and its implications for efficient provision of public goods.

The first section of the paper documents the linkage between the ACS, LEHD and IRS Form 990, and the move to using IRS tax data. Next, we present new descriptive characteristics of nonprofit employment and how such characteristics vary over time. We start by plotting earnings in the nonprofit and for-profit sectors over time. Unconditionally, we find the nonprofit earnings penalty to be 12% relative to for-profit workers. Leveraging workers' transitions between firms over time to estimate worker and firm-specific pay premia in a statistical job-ladder model of earnings (Abowd, Kramarz and Margolis, 1999; Kline, Saggio and Solvsten, 2020), we find most of the penalty to be causal - not driven by selection. We also document large heterogeneity across industries, both in terms of earnings premia/penalties, and worker selection. Lastly, we show that nonprofit and for-profit earnings have been converging over time.

This paper is the first to use comprehensive administrative tax data, covering the near-universe of workers in the US, to quantify and explain the causes of the nonprofit pay differential. Previous studies have estimated nonprofit penalties in specific industries (Weisbrod, 1983; Leete, 2001; Mocan and Tekin, 2003; Hirsch et al., 2018), and documented differences in worker motivations (Francois, 2000; Gregg et al., 2011; Buurman et al., 2012; Mastekaasa, 2020). Much of this work has relied on survey data, however, which can be subject to measurement error as employees sometimes misreport whether they work in the nonprofit sector (Johnston and Johnston, 2019; Bound and Krueger, 1991).

An important exception is Johnston and Johnston (2019), which is the most closely related paper to our work. Johnston and Johnston (2019) use unemployment insurance earnings data from Florida to decompose the nonprofit gap into demand- and supply-side factors. They estimate an average nonprofit earnings differential of -5.5%, with 80 percent of the difference attributable to worker selection, and the remaining 20 percent to demand-side characteristics. Our paper makes several additional contributions. We use the universe of US workers in the tax data over a longer time period, allowing us to capture workers that move across states over time and into major industries that are not covered by unemployment insurance (for example, self-insuring nonprofits and the public sector). In addition to

[1] The Independent Sector (2020)

covering a more comprehensive set of workers, we also observe a richer set of covariates: our data include information on workers' background and household composition, allowing us to precisely examine the effects of socioeconomic background and family dynamics in explaining the nonprofit earnings differential. Understanding the role of these factors is meaningful for our understanding of employment in the nonprofit sector.

The rest of the paper is outlined as follows. Section 2 describes the construction of the data, while Section 3 presents some descriptive statistics. Section 4 outlines our worker-firm job ladder model and Section 5 present the results. Section 6 concludes.

2. Data

2.1 Census Data

The first step in this project was to create a new data linkage between the American Community Survey (ACS), the IRS Form 990, and administration earnings data from the Longitudinal Employer Household Dynamics (LEHD)[2].

The ACS is an annual cross-sectional survey of approximately 3.5 million U.S. households, containing detailed demographic, education and employment information. We use the 2014-2024 ACS survey waves. The IRS Form 990 is an annual information return that all IRS Section 501(c)(3) organizations[3] with gross receipts of over $50,000 are required to file[4]. Information returns contain detailed information on a charity's financials, as well as personnel and program service activities.

The LEHD is administrative data consisting of quarterly earnings by job (employer-employee match) for all employment covered by unemployment insurance (UI). It measures wages and earnings, employer and industry for roughly 96% of all private-sector jobs and most employees of state and local government (Abowd et al., 2009)[5]. Employment in all states and wages in available states are measured on a quarterly basis, and recorded in the Individual Characteristics File (ICF) and Employment History File (EHF), respectively. Using administrative data provides several benefits over survey data, such as larger sample sizes and avoiding concerns about non-random measurement error. Indeed, previous studies have found the size of the non-profit sector to be misrepresented in surveys (Millard and Machin, 2007).

A main aim of this project was to link the LEHD and IRS Form 990 data using employer reference numbers (EINs), adding several new variables to the existing data. Crosswalks between the set of

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[2] Datasets are accessed via a Federal Statistical Research Data Center (FRDC) maintained by the U.S. Census Bureau.
[3] 501(c)(3) nonprofits are those whose mission relates to charity, education, science or public safety testing.
[4] Churches and other houses of worship, and governmental organizations such as public universities are not required.
[5] It excludes employees of the federal government and some private sector employers including independent contractors, the unincorporated self-employed, and certain types of nonprofit employers. The data measure wages and earnings including employers' payroll, workers on paid sick leave and paid vacation.

nonprofit organizations included in the Employer Characteristics File of the LEHD Data and the IRS Form 990 Filings were produced, with filings come from two sources: (i) the NCCS provides data from Form 990 on a sample of 501(c)(3) nonprofit organizations from 1989 through 2015; and (ii) the IRS provides access to the universe of nonprofit organizations which file the Form 990 electronically, for all years between 2009 and 2018. This crosswalk greatly expands the set of characteristics available for these employers. For example, the Form 990 includes information on their revenues and expenditures by source, their use of tax preparers and lobbyists, their total assets and liabilities by the end of each fiscal year, and more.

Crucially for this project, both the LEHD and the Form 990 data include a measure of the number of employees working at each nonprofit. However, in the Form 990, nonprofit organizations are requested to list both the "Total number of individuals employed in the calendar year", and the "Total number of volunteers". The fact that many charities rely heavily on volunteers means that focusing solely on employees (as counted in the LEHD) may result in labelling some charities as small when in fact they are not. Thus, supplementing the LEHD measure of employment with IRS Form 990 data on volunteers helps provide a more accurate measure of charity size. This is important not only for Census records but also for making better-informed government policy. In particular, mischaracterizing charity size, and the implied provision of charitable services, may result in misallocation of government resources.

Linking the LEHD data to IRS Form 990 data via EIN also meant that we were able to compare the LEHD to another more closely examined and value-added data. Specifically, the quality of NAICS and SIC classifications. Any organization that files a Form 990 should be classified as a nonprofit organization in the LEHD data. Thus, by recording the discrepancies between the two data sources, the researchers enhanced the Census Bureau's knowledge of its data collection programs. In particular, IRS Form 990 data are classified according to an alternative system: the National Taxonomy of Exempt Entities (NTEE). The NTEE classifications have been hand-checked for accuracy and quality by the National Center for Charitable Statistics (NCCS). We construct a mapping between the NTEE codes and the NAICS and SIC classifications, and assess whether the NAICS codes assigned to establishments in the LEHD data correspond to the appropriate NTEE code as assigned by the NCCS, where the appropriateness of the match is defined by the mapping. Both the NAICS and SIC classification systems were developed to describe for-profit organizations, but may not align with the NTEE classification used by the IRS for a number of reasons. For example, a nonprofit legal aid organization may be characterized by NAICS under its catch-all nonprofit category, Sector 813, ("Religious, Grantmaking, Civic, Professional, and Similar Organizations"), or it may be considered by NAICS to be part of the legal services industry, Sector 5411 ("Legal Services"). While such a nonprofit may simultaneously be considered a civic organization and a legal services organization, it is more likely to compete with other legal services providers than it is to compete with providers of non-legal civic services. Subsequent research which relies on NAICS codes to identify legal services firms may be incompletely capturing the legal services industry if nonprofit legal services providers are assigned NAICS code 813 rather than NAICS code 5411. Comparison with the NTEE allows us to identify these misclassifications, and to provide a remedy for future researchers.

In this way, comparing measures from independent sources that should be similar, or that should differ in predictable ways, increases the Census Bureau's knowledge of its data collection programs.

2.2 Tax Data

Having worked with the LEHD data for a while, we were fortunate enough to gain access to IRS tax data in 2024. Although the tax data is collected less frequently than the LEHD (annually instead of quarterly), these data brought several advantages. First, it offered us access to the universe of tax records[6]. These anonymized individual records include not only the employer-employee matched records typical of the unemployment insurance records found in the LEHD, but also federal workers, independent contractors and sole proprietors. Second, these data capture information on other earners in each household, as well as demographic data such as age, sex, and geography. For younger workers, we can also observe educational history, parental earnings, and fertility. Having access to this kind of information allows us to explore different potential explanations for nonprofit pay differentials, such as career trajectories, worker productivity, differential tax treatment of non-profit and for-profit employers, family structure, and other non-earnings determinants.

Thus, for these reasons, we pivoted to using the tax data, which we believe enables us to paint a more comprehensive picture of the U.S. nonprofit labor market. Shedding light on the dynamics described above reveals important features of the nonprofit labor market, and how it differs from, and interacts with, the for-profit market.

2.3 Analysis Sample

In order to identify the nonprofit status of workers in the US labor market, we use Forms W-2 and W-3. These provide third-party reported earnings for each worker-employer-year. We use W-3 firm returns, rather than W-2s[7], to identify employer type, with the former containing information on firms' industry and nonprofit status[8]. We then assign W-3 returns to the population of W-2s, where we restrict to full-time, main employer worker-years (Song et al., 2019).

Firm demographics such as industry, geography, income and assets are obtained from Forms 1120, 1065, 990, 1040 Sched. C and 940, and Form 1040 is used to gather information on worker characteristics such as age, gender and household structure.

[6] Unlike the LEHD, which only offered access to a sample of U.S. states, the tax data provides information on the universe of US workers from 1999-2023.

[7] A known issue with W-2s is that firms may use different IDs on a W-2 and an income tax return, meaning that around 15% of W-2s cannot be linked to a business tax filing in that year. This is more common for larger firms, which may have multiple IDs or use a third-party payroll company.

[8] W-3s require firms to use the firm IDs that match the Form W-2's that they issue, and wages to match the sum of W-2 wages.

3. Descriptive statistics

Given that we have data on the universe of US workers, one of the first things we can do is estimate the size of the nonprofit sector – both in terms of the number of workers and payroll. Figure 1 thus presents the nonprofit share of the labor market from 2001 to 2022. On average over the time period, the nonprofit sector makes up just over 10% of all workers, and just under 10% of all payrolls. That is, nonprofits comprise a larger share in terms of the number of workers (rather than payroll), although this difference has converged over time.

Figure 1

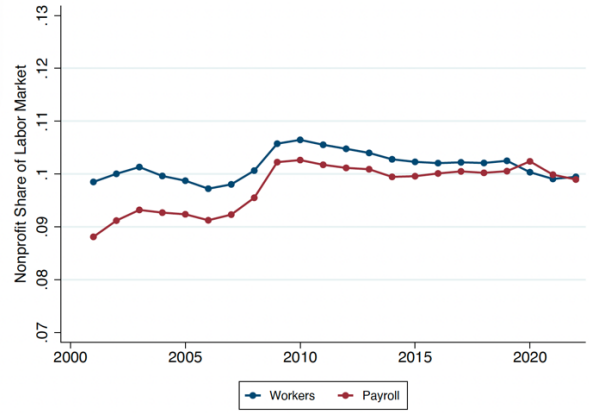
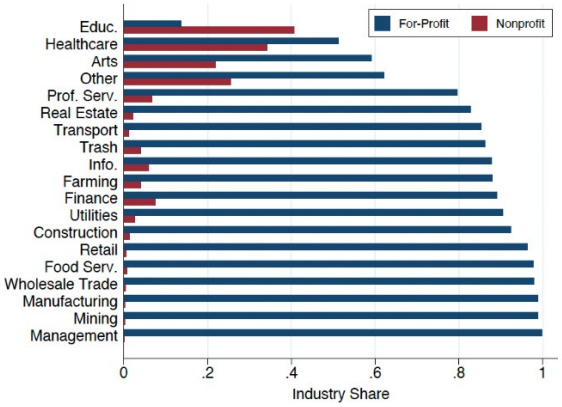


Figure 2



Designating firms as either for-profit or nonprofit also allows us to evaluate the extent of overlap between the sectors. Figure 2 shows that there are relatively few sectors with overlap, with education being the only industry where nonprofits make up a larger share of firms than for-profits (40% versus just over 15%). Unsurprisingly, industries such as manufacturing, mining and wholesale trade have very few nonprofit firms.

3.1 Nonprofit pay premium

Before addressing worker selection, we can also use our data to examine raw differences in average earnings for workers in nonprofit versus for-profit firms. Given compositional differences between the two sectors, it is important to assess heterogeneity in industry-level premia. Here, we restrict attention to industries with overlap. Specifically, we focus on industries (at the 3-digit NAICS code-level) where nonprofit and form-profit firms each account for over 10% of all employment. This means that we exclude sectors such as manufacturing, mining and wholesale trade. Figures 3a-c present within-industry average earnings from 2001 to 2022, with each figure showing the average difference in dark blue but highlighting different sectors. Note that data points to the left of the 45-degree line indicate a nonprofit pay penalty, while points to the right indicate a premium.

Figure 3a

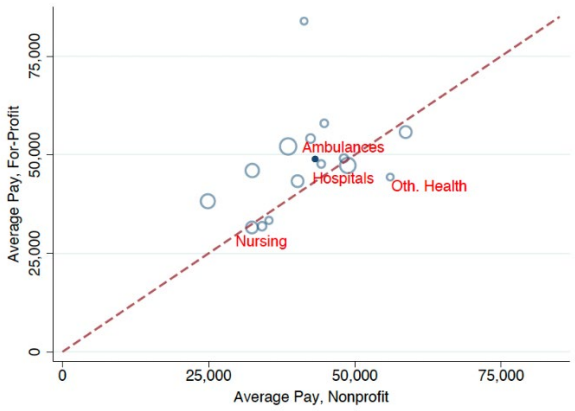


Figure 3b

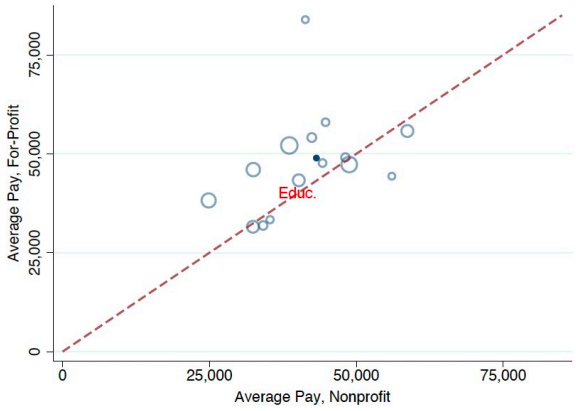
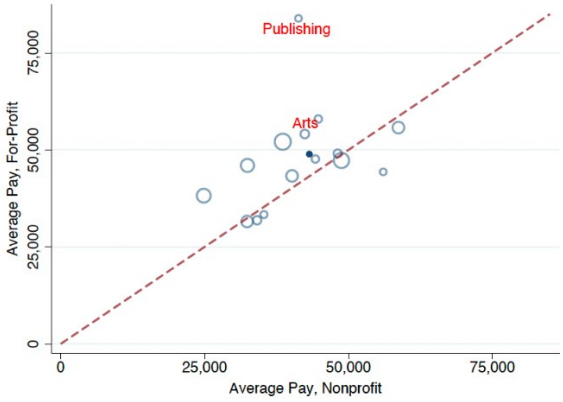


Figure 3c



The dark blue data point shows that, on average, from 2001-2022, pay in the nonprofit sector was lower than in the for-profit sector, indicating a pay penalty. However, there is considerable heterogeneity

across sector. Figures 3a and 3b show that nonprofit pay in the health and education sectors (respectively) is often comparable to the for-profit sector, with some nonprofit health industries even offering a pay premium. This is consistent with evidence that nonprofits operating in sectors with more competition from for-profit firms "act" like their for-profit counterparts (Brown and Slivinski, 2018). Whereas, publishing and the arts exhibit considerable nonprofit pay penalties.

While these raw differentials are informative, the next section addresses how to account for worker selection using an "AKM" worker-firm job ladder model (Abowd et al, 1993; Card et al, 2013; Song et al, 2019)

4. Empirical strategy

We examine the source of nonprofit pay differentials using an AKM model that decomposes the pay gap into worker, firm, and worker-firm match fixed effects (Abowd, Kramarz and Margolis, 1999). Specifically, we estimate the following equation:

$$y_{it} = \alpha_i + \psi_{J(i,t)} + \mathbf{X}'_{it}\beta + \varepsilon_{it} \quad (1)$$

where the dependent variable is wages (y) of individual i at time t. J represents the corresponding firm and ε is the error term. We interpret α as the worker-specific premium, and ψ as the firm-specific premium. ε_{it} captures both transitory earnings variation and idiosyncratic worker-firm match effects. As such, we can compute firm-level earnings premia separate from characteristics of workers at that firm.

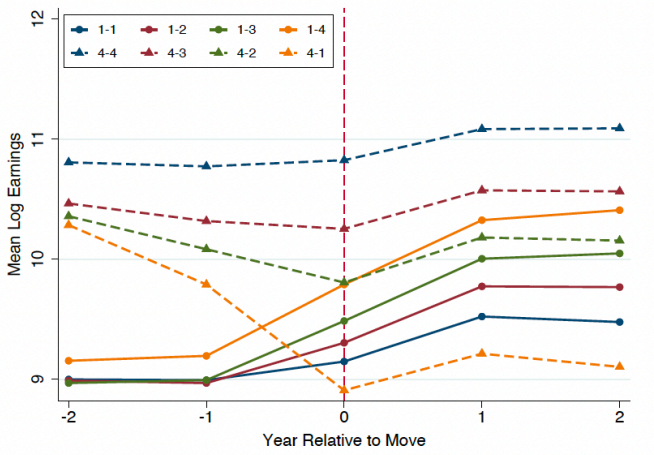
This advantage of this approach is that identification relies on all connected movements by workers between firms. Given that we have access to the universe of US workers, our connected set covers the near-population of full-time workers (see Appendix Table 1). However, in order for this empirical strategy to be valid, the unexplained component of wages can not be systematically correlated with worker-firm types (an "exogenous mobility" assumption). In other words, workers do not move across firms based on the transitory component of wages, for example if mobility is driven by a drop in wages.

One way to test this assumption is to examine a simple event-study of the wage effects of job transitions. Figure 4 presents this event-study, focusing on the two years before and after a job change. To avoid selecting on the job-changer's own wage we classify jobs based on the mean wages of co-workers, omitting the wage of the individual being studied (i.e., using the "leave-out mean" wage). In order for the identifying assumption to hold, it needs to be the case that workers' wage profiles exhibit very little difference in the year before a move. This is largely what we observe in Figure 4, with the exception of moves between the 1st and 4th quartiles: workers moving from a 4th quartile origin firm to a 1st quartile destination firm exhibit small drops in earnings before the move, which could be problematic for our design. However, note that a higher proportion of moves from 4th quartile firms are likely involuntary separations (e.g., mass layoffs) that induce periods of unemployment in the pre-move year, mechanically inducing slightly lower wages in the prior year. The symmetry of the wage losses and gains for those who move between quartile 1 and quartile 4 firms is also important. This suggests that a simple model with

additive worker and establishment effects provides a good approximation of the mean wages resulting from different pairings of workers to establishments.

Lastly, we can use Figure 4 to start to understand the variation in wages. If between-firm wage variation is primarily driven by variation in unobserved worker quality, then transitions between high and low wage firms shouldn't be associated with major wage changes for a given worker. The figure shows that workers moving between quartile 1 jobs experience relatively small wage changes. Whereas, quartile 1 workers moving to jobs with higher paid co-workers experience wage increases, with the largest increases for those who move to quartile 4 jobs. Similarly, there is little change in wages for movers from one quartile 4 to another, but losses when they move to lower quartile jobs, with the largest losses among those moving to quartile 1 jobs.

Figure 4: Job-to-Job Earnings Changes, by Co-Worker Pay



The final assumption to verify is that workers in the same firm receive the same firm-level wage premium. This could be violated if, for example, firms pay larger premiums to high productivity workers. Figure 5a plots the mean residual within deciles of the firm and worker effect distribution, showing that mean residuals are almost always close to zero, especially for high worker effect-high firm effect firms, where one might expect violations. Figure 5b also presents the joint distribution of worker and firm effects, showing it to be relatively bimodal, with most worker-firm observations pairing together low (high) firm effect firms with low (high) worker effect workers.

Figure 5a: Mean Residuals by Worker, Firm Fixed Effects

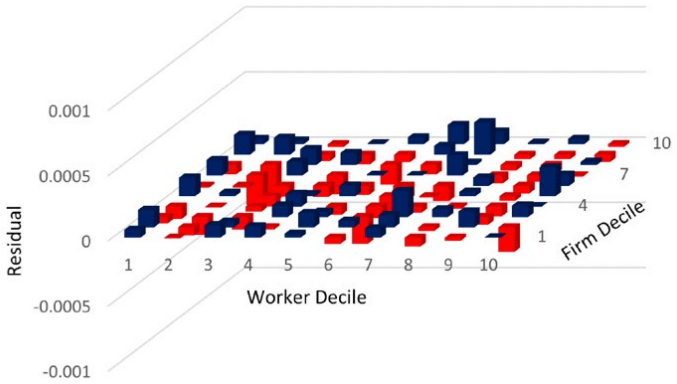
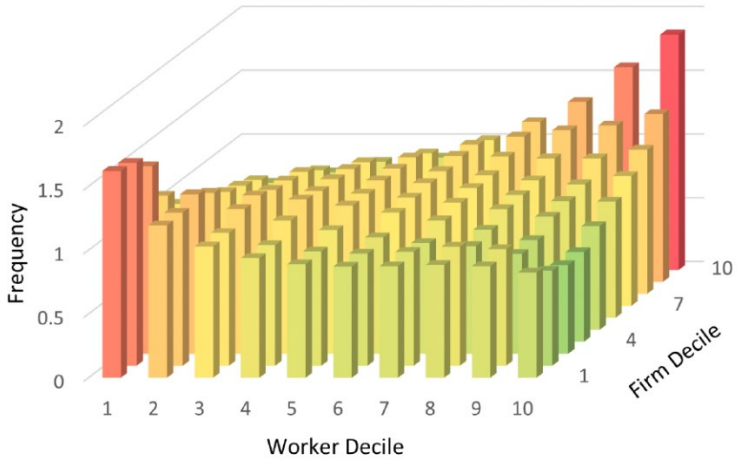


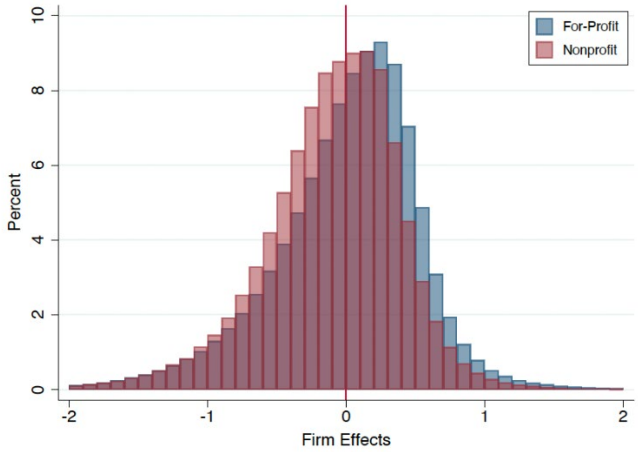
Figure 5b: Joint Distribution of Worker, Firm Fixed Effects



5. Results

Before using the AKM model to examine a selection-corrected measure of nonprofit earnings premia, Figure 6 presents the distribution of firm fixed effects. Recentering around 0, we observe that firm fixed effects are slightly more positive for for-profit versus nonprofit firms.

Figure 6: Distributions of Firm-Level Wage Premia, by Sector



However, in order to disentangle the impact of worker and firm effects on pay premia, we estimate the AKM model specified in equation 1. Using these AKM premia adjusts for differences in worker characteristics within firms, allowing us to test for differential worker selection across for-profit and nonprofit sectors.

Similar to Figures 3a-c, Figures 7a-c present selection-corrected industry premia from 2001 to 2022. Again, each figure shows the average selection-corrected difference in dark blue, but highlight different sectors. Now, data points to the left of the 45-degree line indicate that nonprofit firms have worse workers on average (i.e. negative selection), while points to the right indicate positive selection.

The dark blue data point shows that, on average, from 2001-2022, worker selection plays very little role in pay premia. In other words, on average, workers in the nonprofit sector are no "worse" than workers in the for-profit sector. However, there is again considerable heterogeneity across sector. Figures 7a and 7b show that, on average, workers are of the same quality in the for- versus nonprofit health and education sectors (respectively), and there are even some healthcare sectors where nonprofit workers are positively selected. Whereas, nonprofit workers in publishing are, on average, worse than their for-profit counterparts (Figure 7c). This suggests that the nonprofit pay penalty in publishing documented in Figure 3c is at least partly driven by negative selection. However, the nonprofit pay penalty in the arts is in fact not explained by worker selection.

Figure 7a

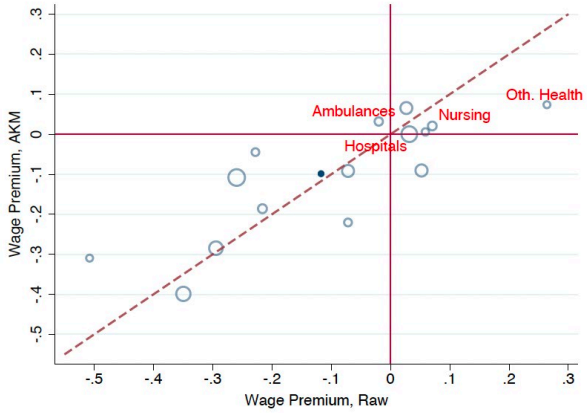


Figure 7b

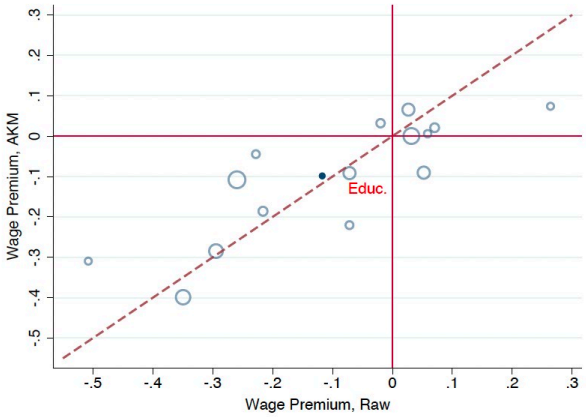
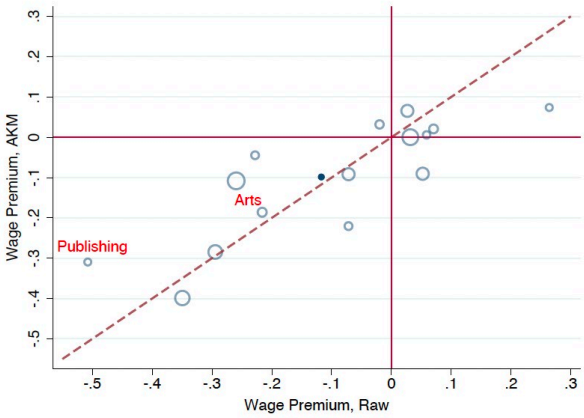
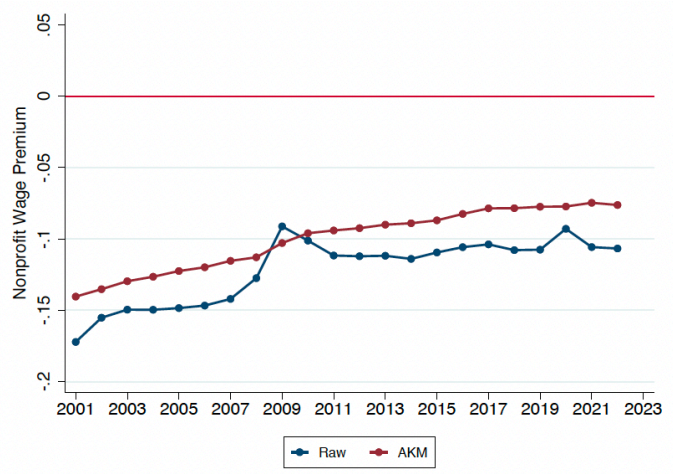


Figure 7c



Lastly, we can examine how pay premia have changed over time. Figure 8 plots the raw vs selection-corrected premia from 2001 to 2022, showing that there has been convergence in nonprofit/for-profit earnings over this time period. In particular, the selection-corrected nonprofit pay penalty narrowed from roughly 14% in 2001, to 7% in 2022.

Figure 8: Raw vs Selection-Corrected Premia Over Time



6. Conclusion and Next Steps

This paper is the first to use population tax data to better understand the US nonprofit labor market, and reassess the nonprofit pay premium. We find, on average, a nonprofit pay penalty of 12%, with most of this differential being causal and not driven by worker selection. However, there exist large heterogeneities across industries, both in terms of pay premia/penalties, and worker selection. We also show that nonprofit and for-profit earnings have converged over time.

Future work will make use of data on household structure to try to explain the persistent nonprofit pay penalty. In particular, we wish to explore the role of career trajectories, tax treatment and family structure.

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Appendix Tables

A.1: Coverage in Estimation Sample vs. Connected Set, All

Level	Estimation Sample	Largest Connected Set
Worker-Years	1,882,594,043	1,866,438,027
Percent of Full Sample	1	.9914
Workers	210,784,644	207,207,843
Percent of Full Sample	1	.9830
Firms	12,682,080	11,004,400
Percent of Full Sample	1	.8677